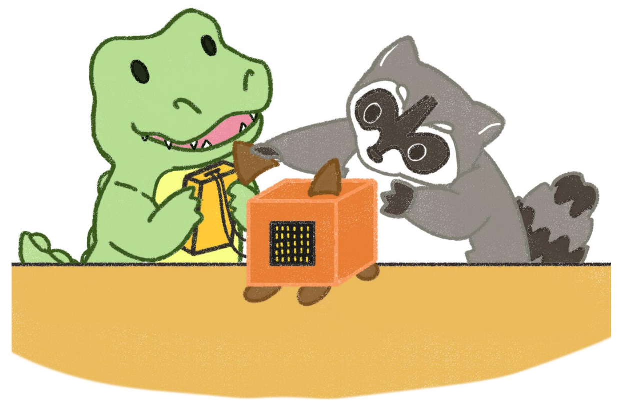

A Scoping Review on STEMM Mentorship in Student Organizations at Institutions of Higher Education

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ENGAGED QUALITY INSTRUCTION THROUGH PROFESSIONAL DEVELOPMENT

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Introduction

This scoping review explored and mapped the existing literature on mentorship within Science, Technology, Engineering, Mathematics, and Medicine (STEMM) student organizations at institutions of higher education. Types of student organizations and mentoring relationships were identified and defined from the literature. Our goal was to promote further investigation and increase visibility of mentorship research in this context.

Methods

Scoping Review

Search Strategy

- We searched for articles containing "student organizations", "mentor", and excluded authors with the name "mentor", across 6 databases:
  - ERIC
  - Inspec
  - Web of Science
  - Scopus
  - Compendex
  - Education Database
- Search results were downloaded as RIS files and uploaded to Covidence.

Title & Abstract Screening, Full Text Screening

- Each article was screened and given a vote of "yes", "no", or "maybe" in Title & Abstract, and in Full Text, either an "exclude" or "include".
- All disagreements between researchers underwent a consensus process.

Table 1. Inclusion criteria for title and abstract screening.

English Language
Journal Article or Conference Paper
Primary Research
Higher Education
Students in Higher Education
Student Organizations
Mentorship

Extraction

Several items were taken from the publications for extraction purposes, including: Student Organization Types, Part of STEMM in the Student Organization, Mentorship Type, and Mentorship Paradigm

Descriptive Analysis

- Data was transferred to an Excel spreadsheet and tabulated, where frequencies of each of the extraction items were calculated.
- Each extraction item was transferred to RStudio and visualized using the programming language R [2].

Results

Scoping Review

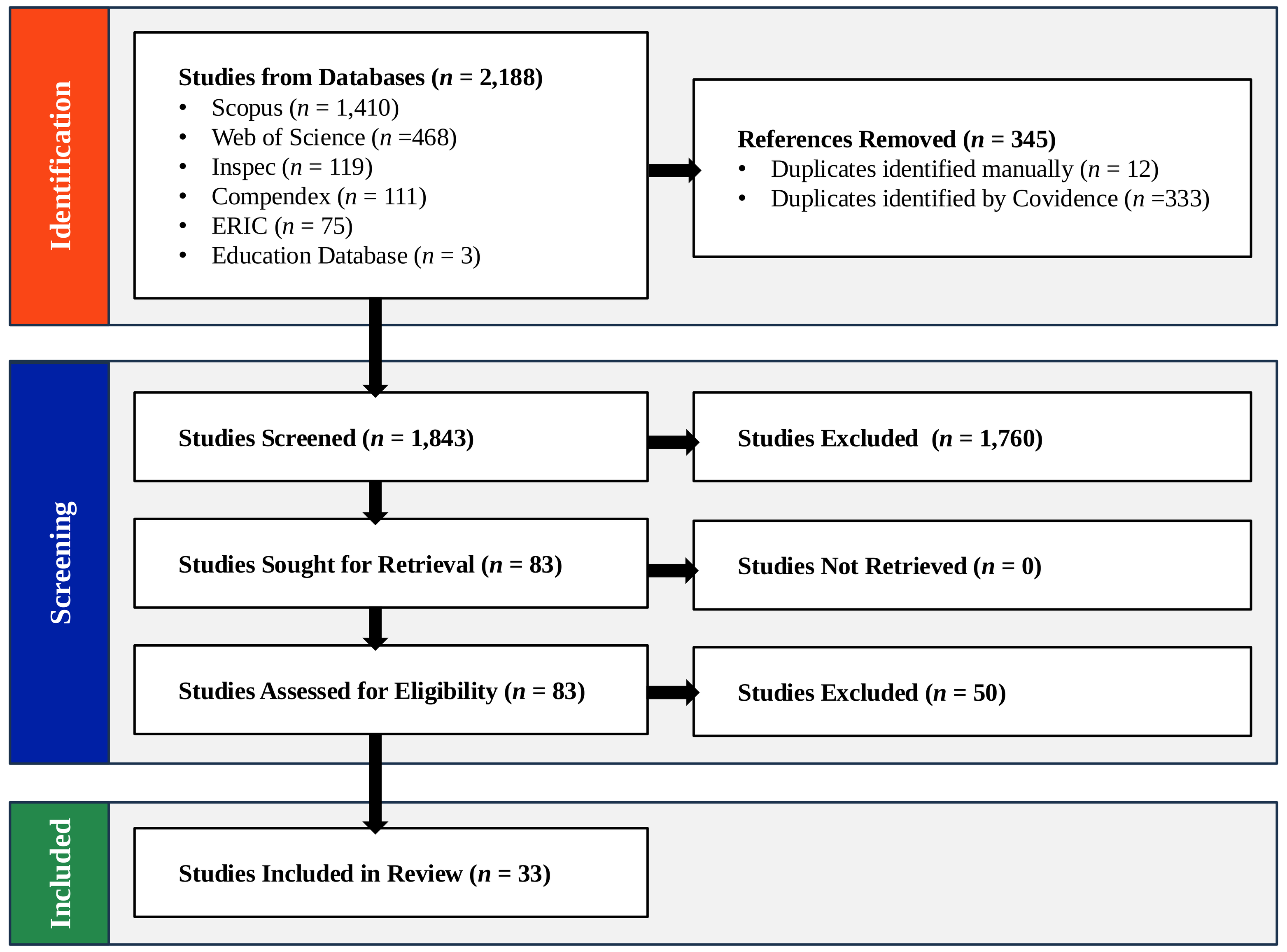
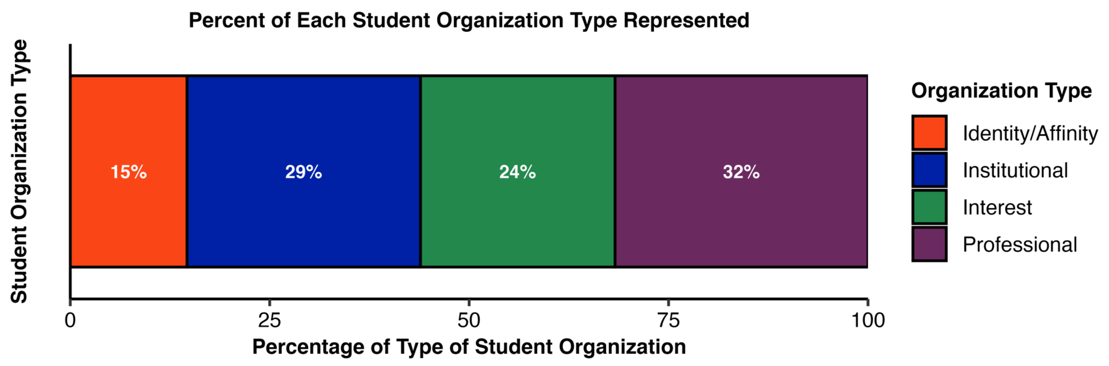


Figure 1: PRISMA Diagram

Database searches resulted in 2,188 publications found. After duplicate removal, 1,843 publications moved into the title and abstract screening. Eighty-three studies moved into the full-text screening, with 33 studies selected for inclusion in the scoping review.

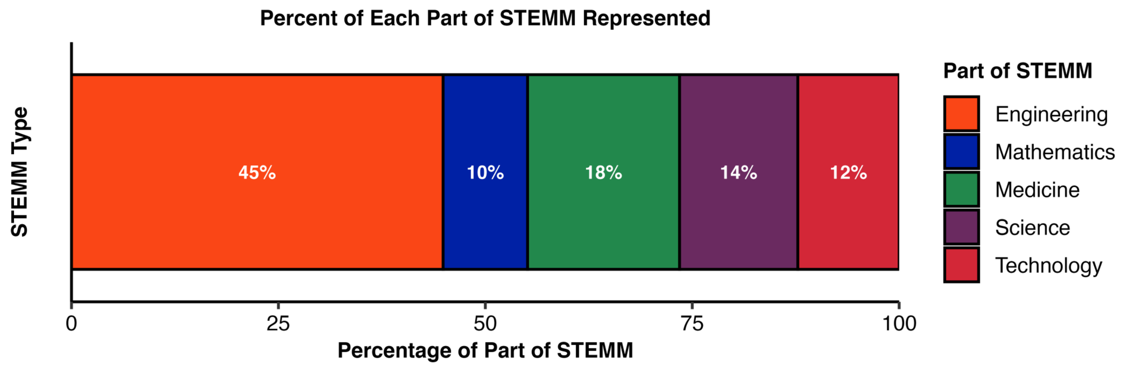
Descriptive Analysis

Figure 2: Student Organization Type



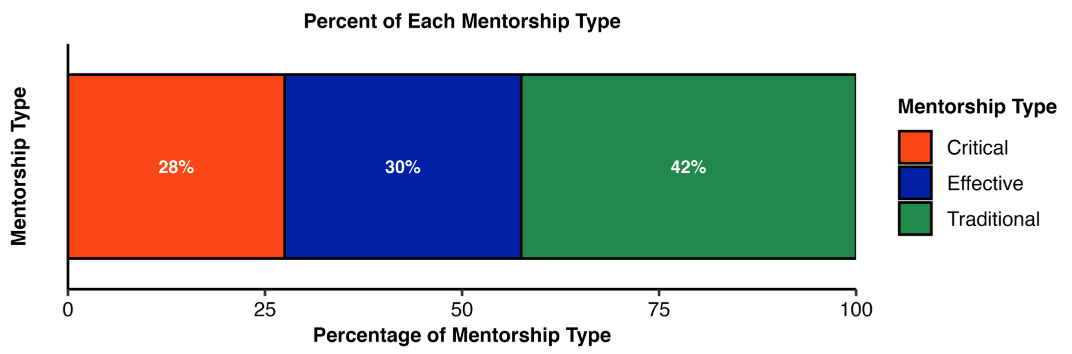
The four types of student organizations include institutional, professional, interest-based, and identity/affinity groups—each defined by academic focus, professional affiliation, shared interests, or common identities.

Figure 3: STEMM Type of Student Organization



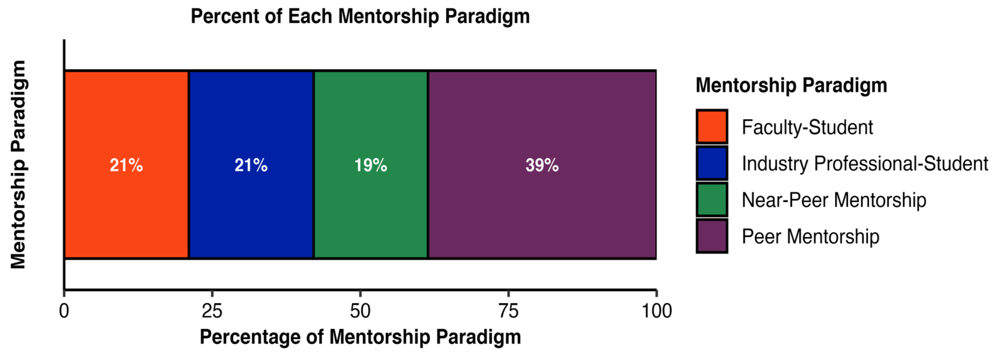
Each student organization was listed by the part of STEMM they were affiliated with; Science, Technology, Engineering, Mathematics, and/or Medicine.

Figure 4: Mentorship Type



Mentorship can be traditional (one-way guidance), effective (mutual exchange), or critical (mutual, with added focus on empowerment, autonomy, and shared identity) [1].

Figure 5: Mentorship Paradigm



Mentorship paradigms reflect who facilitates the relationship: faculty-student, industry professional-student, near-peer (e.g., grad-undergrad), or peer-to-peer.

Conclusion

This scoping review highlights a limited body of research on the topic, with only 33 publications meeting the inclusion criteria. Key findings from the descriptive analysis include:

- Since Compendex and Inspec, engineering-specific databases, were included in the search, it is not surprising that the majority of studies focused on engineering student organizations..
- Professional student organizations were the most common, followed closely by institutional organizations.
- Traditional mentorship emerged as the dominant type of mentorship, while peer mentorship was the most prevalent paradigm.

Next Steps

Building on this scoping review, the research team will continue to explore and develop a codex on the role of mentorship in the STEMM student professional formation. We also aim to raise awareness on mentorship's impact, its benefits, and encourage its broader integration across student organizations. We hope these findings will serve as a catalyst for further research in this important area.

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References

[1] K. Fernandez, K. Chisholm, and N. Ruzycki, "Examining STEMM Mentorship within Student Organizations in Higher Education through a Critical Lens," in 2024 ASEE Annual Conference & Exposition Proceedings, Portland, Oregon: ASEE Conferences, Jun. 2024, p. 47372. doi: 10.18260/1-2--47372.

[2] RStudio Team (2020). RStudio: Integrated Development for R. RStudio, PBC, Boston, MA URL http://www.rstudio.com/.